TOPIC 4 RELATIONSHIP BETWEEN LLN AND SAMPLING DISTRIBUTION

Name: ____________________ Student No.: ____________ Course/Section: ____________ Date: __________

A. CONCEPT CHECK

Answer the following questions.

LEARNING OBJECTIVES

After completing this topic, you should be able to:

1. Explain what the Law of Large Numbers (LLN) states.
2. Describe how the sample mean becomes closer to the population mean as the sample size increases.
3. Explain the effect of different sample sizes on the variability of sample means and the shape of the sampling distribution.
4. Relate the Law of Large Numbers to the sampling distribution.

As sample size increases, the sample mean gets closer to the population mean μ and the sampling distribution becomes narrower.

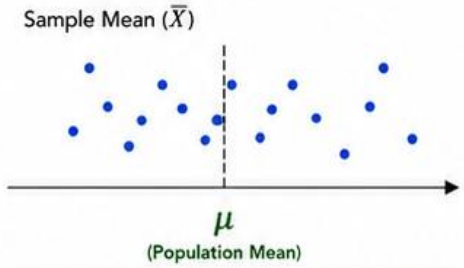
1. What does the Law of Large Numbers (LLN) state? ______________________________
2. As the sample size increases, the sample mean gets closer to which value? ______________________________
3. The symbol for the population mean is __________.
4. The symbol for the sample mean based on n observations is __________.
5. Fill in the blanks to complete the statement of LLN.

As the sample size increases ($n \to \infty$), the sample mean $\overline{X}_n$ tends to the population mean μ.

B. SEE THE IDEA

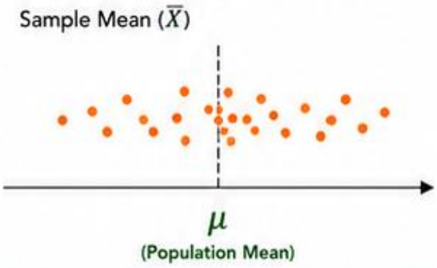
Look at the three situations below. Each dot represents the sample mean from one random sample taken from the same population with true mean μ.

Small Sample Size ($n = 5$)



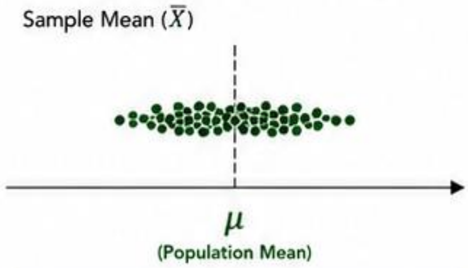
- More fluctuation between samples.
- Sample means are less stable.

Medium Sample Size ($n = 30$)



- Less fluctuation between samples.
- Sample means are more stable.

Large Sample Size ($n = 100$)



- Very little fluctuation between samples.
- Sample means are very stable.

1. Based on the graphs, which sample size gives more scattered sample means? Why?

____

____

2. Which sample size gives sample means that are closer to μ? Why?

____

____

3. What do the graphs suggest about the sample mean as the sample size increases?

____

____

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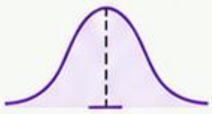
A. UNDERSTAND THE IDEA

Study the sampling distributions of the sample mean below.

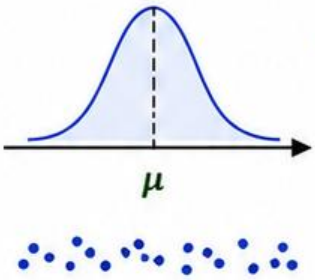


The sampling distribution of $\overline{X}$ shows how sample means are distributed around the population mean μ.

As the sample size increases, the sampling distribution becomes **narrower** and **more concentrated** around μ.

Small Sample Size ($n = 5$)

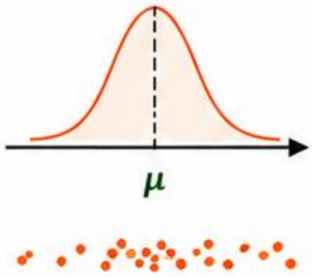
Wider spread (more variability)



- Sample means are **more variable** and **less consistent**.

Medium Sample Size ($n = 30$)

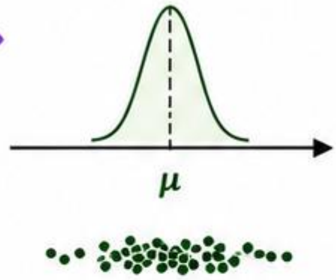
Moderate spread (less variability)



- Sample means are moderately variable and **more consistent**.

Large Sample Size ($n = 100$)

Narrow spread (least variability)

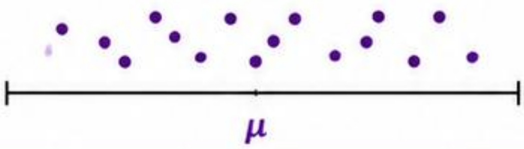


- Sample means are **very consistent** and **very close to** μ.

POPULATION VARIABILITY vs VARIABILITY OF SAMPLE MEANS

Population Variability (σ)

Spread of individual data values in the population.

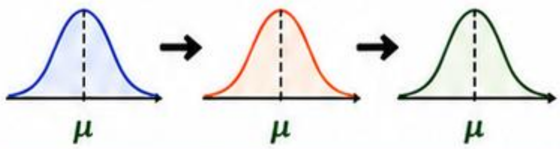


- Does NOT change with sample size.
- It describes the population itself.

VS

Variability of Sample Means ($\sigma_{\overline{X}}$)

Spread of sample means (sampling distribution).



- Decreases as sample size increases.
- It describes the sampling distribution.

B. ORGANIZE YOUR UNDERSTANDING

Fill in the table below using the word bank.

Sample Size (n)	Sampling Distribution of $\overline{X}$	Spread (Variability) of $\overline{X}$	Shape of Sampling Distribution	Position (Center)
5 (small)				
30 (medium)				
100 (large)				

WORD BANK

- Wide
- Moderate
- Narrow
- More variable
- Less variable
- Least variable
- Approximately normal
- Bell-shaped
- Normal
- Centered at μ
- Around μ
- Close to μ

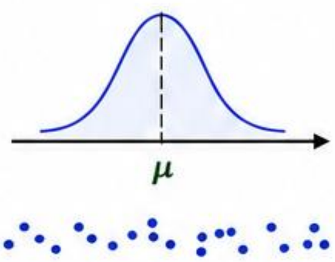
TOPIC 4

RELATIONSHIP BETWEEN LLN AND SAMPLING DISTRIBUTION

C. EFFECT OF INCREASING SAMPLE SIZE

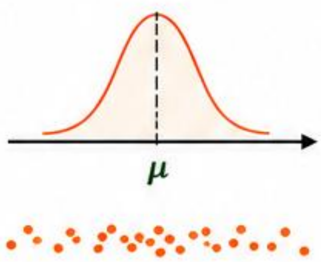
Observe the sequence below and answer the questions.

Small Sample Size (n = 5)



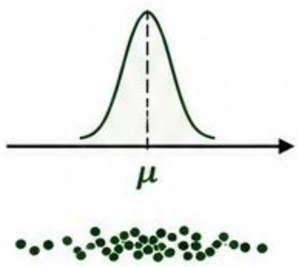
- Sampling distribution of $\overline{X}$ is wide.
- Sample means are more variable and less consistent.

Medium Sample Size (n = 30)



- Sampling distribution of $\overline{X}$ is narrower.
- Sample means are moderately variable and more consistent.

Large Sample Size (n = 100)



- Sampling distribution of $\overline{X}$ is very narrow.
- Sample means are very consistent and very close to μ.

As n increases:

- Variability of sample means decreases.
- Sampling distribution becomes narrower.
- Sample means cluster more closely around μ.
- $\overline{X}$ converges to μ.

D. GUIDED INTERPRETATION

Interpret the situations below.

Situation	What happens to the sampling distribution of $\overline{X}$?	What happens to the variability of $\overline{X}$ ($\sigma_{\overline{X}}$)?	What happens to the closeness of $\overline{X}$ to μ?	Overall Effect
1 Sample size increases from 5 to 30.	______	decreases ______	______	______
2 Sample size increases from 30 to 100.	becomes narrower ______	______	______	estimates become more accurate ______
3 Sample size is very large (e.g., $n \geq 100$).	______	______	sample means become closer to μ ______	______

E. INTERPRET AND CONNECT

Answer the following questions.

1. Why does the sampling distribution of the sample mean become narrower as the sample size increases?

____

____

2. What does the population variability (σ) tell us? What does the variability of sample means ($\sigma_{\overline{X}}$) tell us?

____

____

3. How are the ideas of LLN and the sampling distribution related to each other?

____

____

RELATIONSHIP BETWEEN LLN AND SAMPLING DISTRIBUTION

F. QUICK SUMMARY TABLE

Complete the table to summarize what happens as the sample size increases.

Sample Size (n)	Sampling Distribution of $\overline{X}$	Variability of Sample Means ($\sigma_{\overline{X}}$)	Closeness of $\overline{X}$ to μ	Conclusion
Small (e.g., $n = 5$)		more variable		
Medium (e.g., $n = 30$)	moderately narrow		moderately close to μ	
Large (e.g., $n = 100$ or more)				estimates are very reliable

G. FILL IN THE BLANKS

Fill in the blanks using the correct words from the word bank.

1) As the sample size increases, the sampling distribution of $\overline{X}$ becomes ________________.

2) The variability of the sampling distribution is measured by ________________, and it ________________ as the sample size increases.

3) The sample means become ________________ to the population mean μ.

4) These ideas are supported by the ________________, and together they help us make better estimates about the population.

WORD BANK

narrower

closer

decreases

Law of Large Numbers

$\sigma_{\overline{X}}$

H. THINK AND APPLY

Read each situation and answer the questions.

1 A company wants to estimate the average height of all students in a school. They first measure 10 students and later measure 80 students.

- Which estimate is likely to be more reliable? Why? ____

____

- How are the ideas of sampling distribution and LLN shown in this situation? ____

____

____

2 A researcher is estimating the average number of hours students sleep per night. She takes samples of size 15, then 40, then 120.

- Describe how the sampling distribution of $\overline{X}$ changes as the sample size increases. ____

____

- What happens to the variability of $\overline{X}$? ____
- What happens to the closeness of $\overline{X}$ to μ? ____

____

RELATIONSHIP BETWEEN LLN AND SAMPLING DISTRIBUTION

J. CHALLENGE YOURSELF

Answer the questions.

1 The population standard deviation (σ) is 20. Compare the standard error ($\sigma_{\overline{X}}$) of the sample mean for:

a) $n = 10$

$\sigma_{\overline{X}} =$ ____________

Explain your answer.

____

____

____

b) $n = 25$

$\sigma_{\overline{X}} =$ ____________

Explain your answer.

____

____

____

c) $n = 100$

$\sigma_{\overline{X}} =$ ____________

Explain your answer.

____

____

____

2 If we want the standard error of the sample mean to be no more than 2, and the population standard deviation is 18, what is the minimum sample size needed? Show your calculation.

____

____

____

____

K. REFLECT AND CHECK

Put a ☑ in the box if you can do it.

I can explain the effect of increasing sample size on the sampling distribution.	☐
I can describe how variability of $\overline{X}$ changes as n increases.	☐
I can explain how sample means become closer to μ.	☐
I understand the relationship between LLN and the sampling distribution.	☐
I can apply these ideas to real-world situations.	☐

L. KEY TAKEAWAY

As sample size increases,

- the sampling distribution of $\overline{X}$ becomes narrower,
- the variability of $\overline{X}$ decreases,
- sample means get closer to the true mean μ,
- and our estimates become more reliable.

This is the power of the Law of Large Numbers!